PYTHON FOUNDATIONS - ASSIGNMENT 02

Arithmetic Operators

15 scenario-based Python challenges covering calculations, averages, percentages, both types of division, remainders, powers and mixed arithmetic.

`+` `-` `*` `/` `//` `%` `**` `round()`

Name: _____

Date: _____

Batch: _____

BEGINNER TRACK
15 USE-CASE CHALLENGES

Arithmetic operators - quick reference

Objective: Take user input, convert it to the correct numeric type, perform meaningful calculations, store results in variables, and display clean output. Questions 1-10 focus on the core operators; Questions 11-15 focus on remainder, powers and mixed arithmetic.

+ Addition	**-** Subtraction	***** Multiplication	**/** Normal division
// Floor division	**%** Remainder	****** Power	**round()** Round decimal output

1 Normal division `/`
Use when you need the exact quotient, including decimals.

2 Floor division `//`
Use when you only need the number of complete groups.

3 Remainder `%`
Use when you need to know what is left after equal grouping.

4 Power `**`
Use for squares, cubes and repeated multiplication.

Important: `round()` is a function, not an arithmetic operator. Use it where a division result should be shown with a limited number of decimal places.

Assignment rule: Do not use `if`, loops, lists, dictionaries or functions of your own. Solve using user input, variables, type conversion and arithmetic only.

1 Student Marks, Average & Percentage

`+` `/` `PERCENTAGE` `ROUND()`

A school wants a simple result calculator for three subjects, each marked out of 100.

1. Take student name and marks in Maths, Science and English from the user.
2. Calculate total marks out of 300.
3. Calculate average marks using normal division `/`.
4. Calculate percentage out of 300.
5. Display average and percentage rounded to 2 decimal places.
6. Also calculate how many more marks the student needs to reach a total of 270.
7. Print a clean result summary.

Formula thinking: Percentage = obtained marks / total possible marks x 100.

2 Family Grocery Bill

`+` `-` `*` `/`

A family is buying rice, milk and cooking oil and wants a quick bill summary.

1. Take price and quantity for rice, milk and cooking oil.
2. Calculate the cost of each item using multiplication.
3. Add all item costs to calculate the subtotal.
4. Take a fixed discount amount from the user and subtract it.
5. Add a delivery charge to calculate the final bill.
6. Take the number of family members and calculate the equal cost per person.
7. Round the per-person value to 2 decimal places.

3 Employee Salary Calculator

`+` `-` `*` `/`

An employee wants a simple monthly and yearly salary snapshot.

1. Take employee name, monthly basic salary, transport allowance and food allowance.
2. Calculate gross monthly salary by adding the allowances.
3. Take monthly deductions and subtract them to get net monthly salary.
4. Calculate net yearly salary by multiplying by 12.
5. Calculate the simple average weekly salary by dividing yearly salary by 52.
6. Round the weekly salary to 2 decimal places.
7. Print gross monthly, net monthly, and net yearly salary.

4 Road Trip Fuel & Cost Planner

`*` `/` `+` `ROUND()`

A group is planning a road trip and needs fuel and per-person cost estimates.

1. Take one-way distance, vehicle mileage, fuel price per litre and number of travellers.
2. Calculate round-trip distance.
3. Calculate litres of fuel required using normal division.
4. Calculate total fuel cost.
5. Take total toll and parking charges and add them to the trip cost.
6. Calculate cost per traveller.
7. Round fuel required and cost per traveller to 2 decimal places.

5 Classroom Seating Planner

`//` `*` `-` **COMPLETE GROUPS**

A teacher wants to arrange students into rows with the same number of seats in each full row.

1. Take the total number of students and seats available in one row.
2. Use floor division `//` to calculate how many complete rows can be filled.
3. Calculate how many students are seated in those complete rows.
4. Using subtraction, calculate how many students are left for the incomplete row.
5. Take the number of classrooms and calculate average students per classroom using `/`.
6. Print complete rows, students in complete rows, leftover students and average per classroom.

Think: `//` answers: "How many complete groups can I make?"

6 Restaurant Group Dinner

`+` `*` `/` `//`

Friends are ordering two food items and want to calculate the final bill and payment split.

1. Take price and quantity for two menu items.
2. Calculate each item total and then the food subtotal.
3. Add a packaging charge and service charge entered by the user.
4. Take the number of friends and calculate exact cost per friend using `/`.
5. Also calculate whole rupees per friend using `//`.
6. Calculate the difference between exact share and whole-rupee share.
7. Print all values clearly.

7 Mobile Data Plan Analyzer

`*` `-` `/` `//`

A user wants to understand total data, remaining data and how many complete days it may last.

1. Take daily data allowance in GB and total number of plan days.
2. Calculate the total data available in the plan.
3. Take already-used data and subtract it to calculate remaining data.
4. Take expected daily usage and calculate the exact number of days remaining data can support using `/`.
5. Use `//` to calculate only the number of complete days.
6. Take plan price and calculate average price per day.
7. Round decimal outputs to 2 places.

8 School Trip Budget

`+` `*` `/` `//`

A teacher needs a simple budget for a school trip.

1. Take number of students, bus cost, entry fee per student and meal cost per student.
2. Calculate total entry-fee cost.
3. Calculate total meal cost.
4. Add all costs to get total trip cost.
5. Calculate exact equal cost per student using `/`.
6. Take maximum students per bus and use `//` to calculate the number of completely filled buses.
7. Print the complete budget and rounded per-student cost.

9 Monthly Savings Planner

`+` `-` `*` `/`

A person wants to estimate monthly savings and yearly savings using simple arithmetic only.

1. Take monthly salary and additional monthly income.
2. Take rent, food, travel and other monthly expenses.
3. Calculate total monthly income and total monthly expenses.
4. Subtract expenses from income to calculate monthly savings.
5. Calculate yearly savings by multiplying monthly savings by 12.
6. Calculate the percentage of monthly income being saved.
7. Round the saving percentage to 2 decimal places.

10 Time Conversion & Study Schedule

`//` `*` `-` `/`

A student records total study time in minutes and wants to understand it in different forms.

1. Take total study time in minutes.
2. Use `// 60` to calculate complete study hours.
3. Calculate minutes used by those complete hours using multiplication.
4. Use subtraction to find the remaining minutes.
5. Take number of subjects studied and calculate average minutes per subject using `/`.
6. Take number of study days and calculate average minutes per day.
7. Print the time as complete hours + remaining minutes, and round averages to 2 decimals.

Part B - Remainder, Power & Mixed Arithmetic

The remaining arithmetic operators are **%** (modulo/remainder) and ****** (exponent/power). These questions also mix them with the operators you already used.

11 Candy Distribution

`//` `%` EQUAL GROUPING

A teacher has a box of candies and wants to distribute them equally among students.

1. Take total candies and number of students.
2. Use `//` to calculate how many candies each student gets completely.
3. Use `%` to calculate how many candies remain after equal distribution.
4. Take the number of extra candy boxes and candies in each box.
5. Add the extra candies to the original total.
6. Repeat the equal-share and remainder calculations for the new total.

Key idea: `//` gives the complete share; `%` gives what is left.

12 Square & Cube Calculator

`**` SQUARE CUBE

A student wants to explore powers using real measurements.

1. Take a number from the user and calculate its square using `** 2`.
2. Calculate its cube using `** 3`.
3. Take the side length of a square and calculate its area using power.
4. Take the side length of a cube and calculate its volume using power.
5. Calculate the difference between the cube of the original number and its square.
6. Print all results with meaningful labels.

13 Packaging Factory: Boxes & Leftovers

`//` `%` `*` `+`

A factory packs identical items into boxes of equal capacity.

1. Take total number of items and capacity of one box.
2. Calculate the number of completely filled boxes using `//`.
3. Calculate unpacked leftover items using `%`.
4. Calculate how many items are inside the completely filled boxes.
5. Take the number of new items arriving and add them to the original total.
6. Calculate the new number of full boxes and new remainder.
7. Print both the original and updated packaging summary.

14 Game Score Power-Up

`**` `+` `-` `*`

A simple game calculates a bonus score using a player's level and a power formula.

1. Take player name, base score and player level.
2. Calculate level power as `level ** 2`.
3. Take bonus points per level and multiply them by the player's level.
4. Add base score, level power and level bonus to get the total score.
5. Take penalty points and subtract them from the total.
6. Calculate the difference between the final score and the original base score.
7. Print a complete score summary.

15 School Carnival - Master Arithmetic Challenge

`+` `-` `*` `/` `//` `%` `**`

A school carnival wants one program that combines everything learned in this assignment.

1. Take number of students, ticket price per student and total food-stall income.
2. Calculate total ticket income and total carnival income.
3. Take decoration cost, venue cost and food expense; calculate total expense.
4. Subtract total expense from total income to get the remaining amount.
5. Calculate remaining amount per student using normal division `/`.
6. Suppose prize packs contain a fixed number of prizes. Use `//` to find complete packs and `%` to find leftover prizes.
7. Take a game level number and calculate `level ** 2` as a bonus-score value.
8. Round decimal money values to 2 places and print a complete carnival summary.

Challenge: Before running your program, write beside every calculation which arithmetic operator you expect to use and why.

Submission checklist

- I converted numeric input using `int()` or `float()`.
- I used `/` when I needed an exact division result.
- I used `//` when I needed complete groups only.
- I used `%` when I needed the remainder.
- I used `**` for square/cube or power calculations.
- I used meaningful variable names.
- I stored important calculations in separate variables.
- I used `round()` only to format decimal results where needed.
- I tested at least two different input sets.
- I did not use `if`, loops, lists or functions.

Operator set completed: `+`, `-`, `*`, `/`, `//`, `%`, `**`. These are the standard Python arithmetic operators you need at this stage.

Assignment Complete

Questions attempted: _____ / 15 Teacher review: _____